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# Enhancing thermal-moisture comfort of a chest wader with lining for fishermen

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**Abstract**

As basic protective clothing, a chest wader needs to be worn by fishermen during wading work. Its thermal and moisture performance would directly influence fishermen's comfort and health. In order to increase the thermal-moisture comfort for fishermen, this research aimed to design a new lining chest wader and to carry out both subjective and objective evaluations on its comfort properties. By simulating moisture absorption under a hydraulic pressure environment, we tested the water absorption and conductivity of the selected fabrics and water-absorbent resin materials, and used an appropriate combination of the fabric and the resin to cover different sweating areas of the human body. Ten subjects were recruited to conduct the wearing test on the chest wader to evaluate its thermal-moisture performance. The objective evaluation involved tracking the changes in temperature and relative humidity at four body parts when a participant wore the chest wader, while the subjective evaluation included a questionnaire about the wearing experience on the thermal-moisture comfort and operation convenience. The experimental results showed that the new lining enhanced the comfort significantly. Compared with the common chest wader, the new lining reduced the relative humidity of chest wader on average by 30% approximately, with the largest improvement being at the calf, followed by the thigh and crotch. The thermal-moisture performance of the chest wader was also influenced by the hydraulic pressure of the working environment and the ease allowance to the wearing body.

**Keywords**

fishermen, chest wader, hybrid lining, thermal-moisture comfort

Fishing has been listed as one of the most dangerous industries by the International Labor Organization, due to frequent accidents often threaten fishermen's lives.[1,2] In addition, the working environment is adverse because of high humidity and severe temperature. As basic protective clothing, a chest wader forms a sealed microclimate space for fishermen to perform underwater operations, and therefore should have a good waterproof property at certain hydraulic pressures. Meanwhile, the aqueous vapor and sweat generated from fishermen's activities must be discharged in a timely manner, because a long duration in a highly humid and imperious environment could cause tremendous discomfort, arthritis and other diseases in the lower limbs. A previous survey indicated that more than 90% of the surveyed fishermen suffered from arthritis after three years of service.[3]

Most chest waders on the market are constructed of two layers: an outside layer that is made of a waterproof fabric (such as fabric with rubber or polyvinyl chloride (PVC) coating) and an internal layer that is lined with a mesh fabric. A fisherman often wears a chest wader with underwear, which helps to adjust the moisture in the microclimate inside the chest wader until it is saturated. Currently, the existing lining used in a wader is only functional for dredging water, and has no effect on moisture absorption.

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Although there are relevant ergonomic evaluation standards for personal protective equipment (PPE), such as EN ISO 20344:2012, Directive89/686EEC,[4] the evaluation method differs among different protective equipment products. In terms of functionality, protective footwear has a high similarity to the chest wader, since both contain barrier materials that prevent heat and moisture from dispatching to the outside in a timely manner and the internal relative humidity (RH) can reach 100%.[5,6] The proper design of a combined lining can improve the moisture absorption performance of protective footwear whose outside layer is made of rubber.[7,8] Through thermal-moisture comfort analysis and feedback from footwear customers, the optimal fabric selection for shoes can be determined.[9]

To meet the needs for both protection and comfort in protective clothing, the multi-layer combination method has been adopted in the design of the firefighter uniform.[10] Because a large amount of heat, sweat and aqueous vapor could be generated by a firefighter's body during operation, the moisture absorption and conductivity of the fabric close to the skin is the key factor to the thermal-moisture comfort of the firefighter's uniform. Regardless of the number of layers in the structure, the physical property of the underwear and lining plays the most significant role in moisture management. The lining can obviously influence the sweat distribution and the final perception of the skin[11], and in turn, moisture management can greatly change the humidity and temperature of the microclimate formed by the protective clothing.[12]

Karlsson and Rosenblad[13] examined the uncomfortable factors and requirements for fishing operations, and designed new work clothes by improving several component structures. Lucas et al.[14] studied the wearing comfort of different personal flotation devices and Mirka et al.[15] analyzed the low back pressure of fishermen when cultivating crabs. However, research on a chest wader has been lacking. Without a functional design for humidity management, a chest wader fails to be an effective device for preventing the occupational diseases of fishermen as intended. Poor wearing comfort has undermined fishermen's willingness to use this PPE.[16]

This project aimed to improve the microclimate of the chest wader in underwater operations by increasing the lining's capacity for moisture absorption, and to design a new lining chest wader with a fabric combination and high water-absorbent resin materials. In order to select appropriate fabrics, moisture absorption and conductivity experiments were conducted to simulate the hydraulic pressure and to test the moisture transfer performance of the new lining and water-absorbent resin materials. According to the sweating characteristics of body areas, the lining design applied different fabrics to different areas to form a hybrid structure. The chest wader operation test was then conducted in a chosen fishery, and 10 subjects took part in the outdoor experiments. The comfort evaluations on the chest wader were performed both objectively, with the temperature and RH data at four parts of a human body, and subjectively, with the subjects' feedback from a specific questionnaire survey.

## Materials and methods

### *Fabric test*

Lining fabrics used in the chest wader must have good moisture performance. For fishermen working on land, the lining performance can be evaluated by different standard physical property tests. However, the hydraulic pressure of the underwater environment may alter the lining's performance. The fabric test was to investigate the moisture absorption and transfer ability of the lining both on land and underwater. An underwater pressure simulation test was designed to evaluate the vapor permeability, moisture conductivity and hygroscopicity of the lining in an underwater environment. The lining should not only meet the requirements of moisture absorption and transfer, but also ensure operational convenience with a chest wader. Six knitted fabrics with high absorption and quick-dry ability were selected from the commercial market as the samples. The specification parameters of these fabrics are shown in Table 1.

A superabsorbent polymer, painted with sodium polyacrylate granules (formula $(C3H3NaO2)_n$), was adopted to make the lining combination with fabrics for the water-absorbent resin materials. The superabsorbent polymer can absorb water hundreds times of its own weight and has good water retention under a pressure.[17]

The hygroscopicity, moisture permeability and moisture conductivity of the fabrics were obtained from the standards GB/T9995-1997, FZT01071-2008 and ISO15496:2004, respectively. The four underwater pressures were simulated by applying identical weights to the fabric. According to the relationship between the hydraulic pressure and the gravitation force (weight), the weight can be calculated as

$$w = \rho\, h\, S \tag{1}$$

where $\rho$ is the water density, $h$ is the water height and $S$ is the action area. When $S$ is $10 \times 10\text{cm}^2$, the weights corresponding to the hydraulic pressures at water depths of 0.4, 0.7, 1.0 and 1.3 m are 4, 7, 10 and 13 kg, respectively.

The six fabrics and the superabsorbent polymer film were cut into $10 \times 10\text{cm}^2$. The superabsorbent polymer

**Table 1.** Specification parameters of the six fabrics.

| Fabric code | Composition | Structure | Weight ($g/m^2$) | Thickness (mm) | Wale density (loops/5 cm) | Weft density (loops/5 cm) |
|---|---|---|---|---|---|---|
| F1 | 100% Polyester (Coolmax®) | Warp knitted mesh | 130 | 0.520 | 98.0 | 170.0 |
| F2 | 100% Polyester (Cooldry®) | Warp knitted mesh | 149 | 0.629 | 95.0 | 188.0 |
| F3 | 60% Cotton 40% polyester | Laying-in stitch | 238 | 1.141 | 54.0 | 97.0 |
| F4 | 100% Polyester | Purl | 145 | 0.349 | 103.8 | 171.8 |
| F5 | 100% Polyester | Warp knitted mesh | 166 | 0.481 | 105.8 | 160.2 |
| F6 | 88% Polyester (Coolpass®)/ 12% polyester | Single jersey | 121 | 0.378 | 100.0 | 134.2 |

Note: Coolmax®, Cooldry® and Coolpass® are registered trademarks of profiled polyester fibers.

film was a non-woven fabric painted with sodium polyacrylate granules, and has the weight of $6.5 \pm 0.15$ g at $10 \times 10cm^2$. The sodium polyacrylate granules were painted evenly at a density greater than 0.1 g per $10 \times 10cm^2$.

All the samples were conditioned at a temperature of $20 \pm 2°C$ and $65 \pm 3\%$ RH for 24 hours. The fabric test was carried out in the following steps:

1. record the weight of the fabric sample and the superabsorbent polymer film;
2. drop $4 \pm 0.1$ g distilled water on the fabric sample and record the weight;
3. attach the superabsorbent polymer between the fabric back side and the waterproof membrane to form a combination, and put a tagger on the combination to cover the whole sample;
4. put four different weights on the tagger for 3 and 10 minutes, respectively, and record the weight changes of the fabric sample and the superabsorbent polymer film;
5. repeat the above steps three times and calculate the average of the readings in the three tests.

The two time durations used in the fourth step were decided by a pre-test on the fabric, which showed that the water transfer was still active within 3 minutes, and almost stopped after 10 minutes. However, the water evaporation of the fabric increased over time, which could influence the measurement of the transferred water.

## Design of the hybrid lining

*Size specification.* A chest wader, in the popular jumpsuit style, was chosen as the test sample from a labor protection supplies store. Its outer shell was made of rubber, and its size specification is shown in Figure 1(a). To allow the wearer to move conveniently in the water, the ease allowance of the chest wader, especially at the girths, was chosen to be larger than a regular one. It had the same chest girth, waist girth and hip girth (128 cm). The pattern of the hybrid lining was designed based on the chest wader (Figure 1(b)), which kept the upper hem, chest, hip, thigh and calf the same as those of the chest wader. To avoid the fabric stack at the crotch, the length from chest to crotch in the lining was decreased by 2 cm. When the subjects wore the lining chest wader, the ease allowances were $42 \pm 6$ cm at the chest, $36 \pm 8$ cm at the hip, $31 \pm 5$ cm at the thigh and $7 \pm 3$ cm at the calf.

*Fabric composition.* The new lining was designed to deal with perspiration efficiently based on the body mapping of sweating patterns.[18] A body covered by a chest wader was divided into two parts. One was the region with high sweat rates and high sweat retention, such as the back, waist to hip and calf. The other was the normal sweat rate region, that is, chest and thigh. Figure 2 shows the design of the new lining with a hybrid structure.

The high moisture transfer fabric and the superabsorbent polymer film were used in the first region to supply high water absorption capacity, as shown in the shaded areas in Figure 2. High hygroscopicity performance fabric was used in the second region as a lining. The superabsorbent polymer film was put in a pocket attached to the lining.

The pocket consisted of two overlapped pieces to guarantee that the film was anti-slip and replaceable, which made the film contact with the lining closely. The water could be transferred efficiently from the fabric to the water-absorbent resin. The blue strap is the Velcro to fasten the lining inside the chest wader. The fabric with the best performance of moisture transfer in the fabric test was chosen for the first region, and that with the best hygroscopicity was chosen for the other.

## Comfort evaluation

To acquire a comprehensive and reliable comfort assessment of the new lining, this study involved both

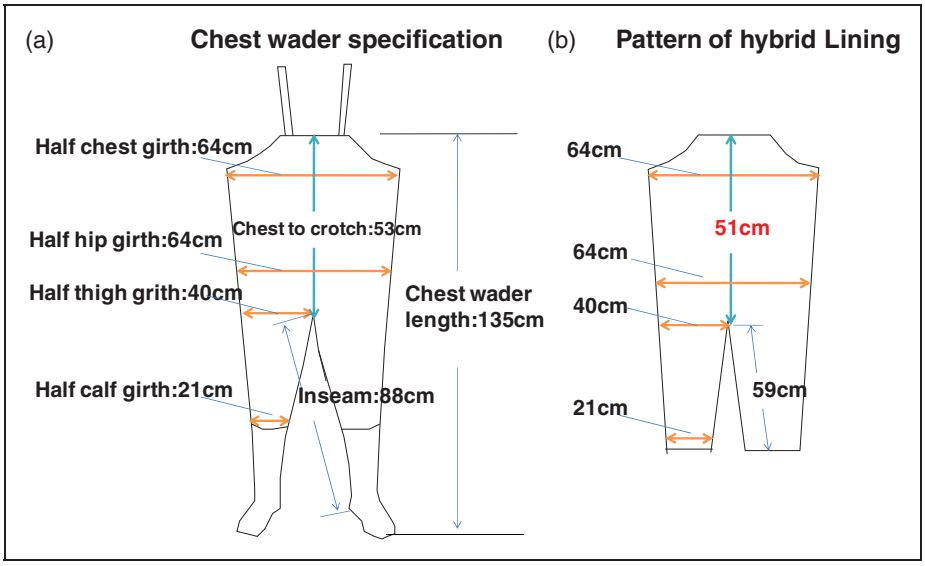


**Figure 1.** Chest wader specification and pattern of the hybrid lining.

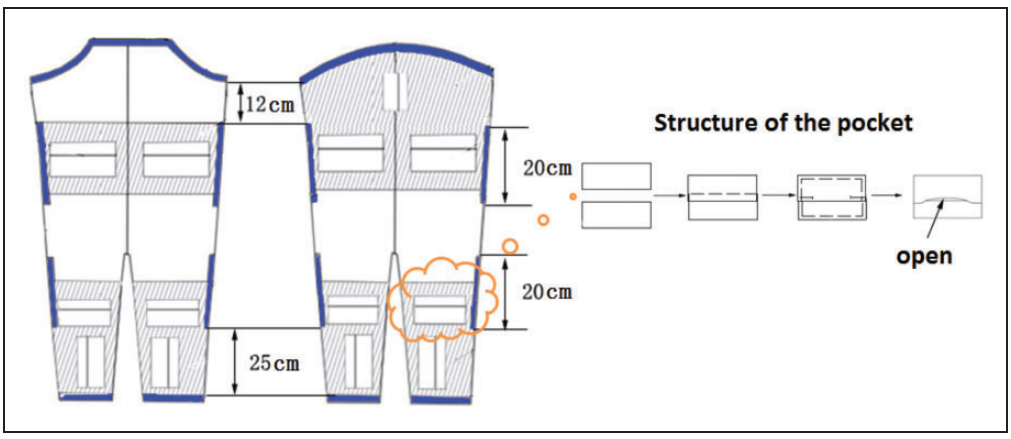


**Figure 2.** Design of the hybrid lining. (Color online only.)

objective and subjective evaluation methods. The subjective assessment was performed with a questionnaire, and the objective assessment was conducted by recording temperature and RH data at four lining locations.

*Participants: field test environment.* Ten subjects (three females and seven males, aged $22 \pm 2$ y, healthy) were recruited in this study. The field test was performed in an outdoor pool that simulated a fishery. Intraday climate parameters, including temperature, RH, illumination intensity and wind and subjects' information are shown as Table 2.

Each subject was given a two-piece outfit consisting of a fitted T-shirt and a pair of trousers made of 100% cotton. The underwear specification was maintained the same for all subjects to avoid the influence in the test by the different underwear cover. Throughout the entire test, subjects were kept uninformed about the difference between the original lining and the new lining.

*Experimental procedure.* The experimental procedure was designed according to EN ISO 20344:2011 and fishermen's work flow. However, our test involved 10 subjects, which was more than the number of subjects (three) required by the standard. The experimental procedures tried to simulate typical tasks undertaken in general use. Body exercises in the test were performed both in water and on the ground by simulating activities and body motions originated from necessary and typical physical work in the fishery. As illustrated in

**Table 2.** Subjects and test environment.

| Gender | Body height (cm) | Body weight (kg) | Water Temperature (°C) | Air temperature (°C) | Illumination intensity (×1000 lux) | Wind velocity (m/s) | Relative humidity (%) |
|---|---|---|---|---|---|---|---|
| Female | 167 | 53.4 | 17.7 | 35.6 | 74 | 1.2 | 58% |
| Male | 172 | 57.6 | 17.9 | 36.8 | 79.6 | 1.1 | 52% |
| Male | 177 | 60.0 | 18.7 | 37.8 | 81 | 1.6 | 51% |
| Male | 173 | 62.1 | 18.9 | 38.6 | 85 | 1.3 | 47% |
| Female | 164 | 53.4 | 19.0 | 38.9 | 87.7 | 1.0 | 44% |
| Male | 172 | 57.2 | 19.3 | 39.0 | 90.8 | 1.2 | 43% |
| Male | 167 | 65.3 | 20.1 | 41.1 | 100.6 | 1.1 | 39% |
| Male | 171 | 60.0 | 20.2 | 41.3 | 101.2 | 2.0 | 37% |
| Female | 166 | 54.3 | 20.8 | 41.5 | 103.8 | 1.5 | 38% |
| Male | 170 | 58.5 | 20.6 | 41.4 | 102.6 | 1.6 | 39% |

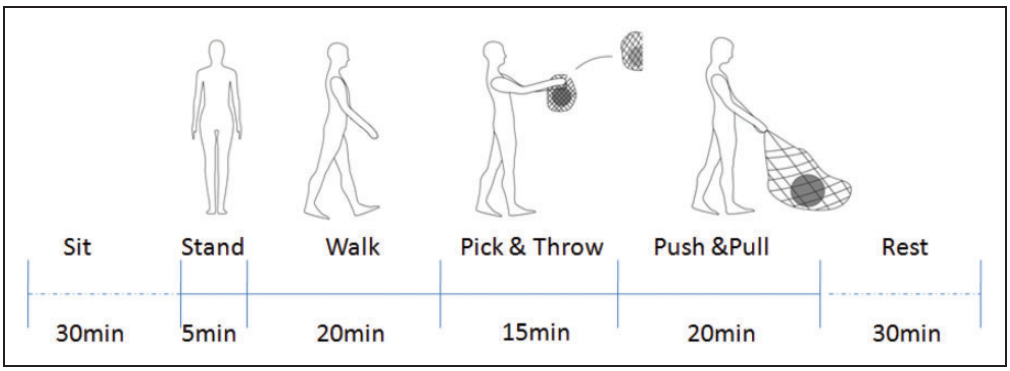


**Figure 3.** Exercise simulating fishing motions.

Figure 3, the evaluation test consisted of physical activities by following the procedures below:

1. sit quietly for 30 minutes before the beginning of the test;
2. stand in water for 5 minutes;
3. walk at a speed of 3 km/h in water for 10 minutes and at a speed of 5 km/s on the ground for 10 minutes;
4. pick up a fake fish with a weight of 1 kg, and throw it in a box that was 1 m away in water, and repeat this 20 times (about 15 minutes);
5. push and pull the net cage, which was about 30 kg, for 20 minutes at a speed of 1–2 km/h in water;
6. rest for 30 minutes on the ground to prepare for the next test.

The 10 subjects wore a common chest wader and the new lining chest wader, respectively, and sat to rest for 30 minutes before each test.

The temperature sensors (NTC, MF52B103F, Hangzhou Loggertech Co., Ltd) and RH sensors (Honeywell HIH5030) were placed at the back (middle), inside of thigh (right leg), crotch (middle) and inside of calf (right leg) to monitor changes of the microenvironment, as shown in Figure 4. The temperature and RH data on the sensors were recorded at 5-second intervals.

Every volunteer performed the subjective evaluations in the questionnaire at the end of each test. The questionnaire was about the body's thermal-moisture comfort and operational convenience when wearing a chest wader, namely descriptions of thermal comfort, moisture, permeability, adherence to body, stiffness and overall comfort. Each subject was asked to give a score on the seven sensations for each question, as shown in Table 3.

At the same time, the subjects were asked to indicate which body part among the waist, hip, crotch, thigh, calf, knee, back and chest yielded the most sweat or most moisture. After finishing the tests with two chest waders, the subjects were asked about in which body part they felt the largest difference when wearing the two chest waders.

## Result and discussion

### *Moisture permeability tests*

*Fabric test.* Table 4 presents the results of the fabric hygroscopicity, moisture conductivity and moisture permeability tests. The results indicated that the humidity absorption performance of the six fabrics was in the following order: F4 < F2 < F5 < F6 < F1 < F3. Fabric F3 was a polyester cotton blended fabric that had the largest moisture regain and the best hygroscopicity. The moisture permeability test demonstrated that the amount of water vapor transferred on the front side was slightly larger than that on the back side, and the moisture permeability on the front side was in sequence from high to low: F6 < F5 < F3 < F2 < F4 < F1. It was observed that the amount of transferred water vapor was influenced by the fabric structure, density and tightness. Fabric F1 had a four pipeline fiber morphology and a large interval between fibers, which made the moisture transmit quickly through the fabric and resulted in good moisture permeability. Fabric F4 had the smallest thickness and largest inter-yarn spaces, which also granted it good moisture permeability. Through the moisture conductivity test, it was found that the order of the six fabrics for the wicking height was F5 < F2 < F3 < F4 < F1 < F6. This is because fabrics F1, F4 and F6 had large yarn intervals and surface areas permitted good capillary effects and strong moisture conductivity. On the other hand, the remaining fabrics had regular structural parameters, and thus relatively weak moisture conductivity.

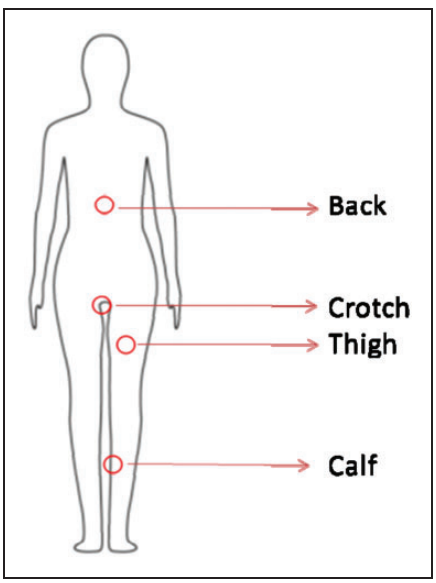


**Figure 4.** Placement of temperature and humidity sensors.

*Lining test.* The water transfer test on the lining materials was performed under simulated hydraulic pressures (Table 5). The water loss rates of the resin materials in all lining combinations increased as time went on and the hydraulic pressure was enhanced (i.e. water depth). The water loss rate of fabric F3 lining combination showed the largest change (from 15.6% to 31.94%) when the water depth varied from 0.4 to

**Table 4.** Fabric test results.

| Fabric | Hygroscopicity (%) | Moisture conductivity (cm) | Moisture permeability (g/(m$^2$·d)) | |
|---|---|---|---|---|
| | | | Front side | Back side |
| F1 | 1.05 | 18.2 | 12,497.5 | 12,040.2 |
| F2 | 0.62 | 13.6 | 8924.1 | 8446.0 |
| F3 | 4.72 | 14.3 | 7169.6 | 6639.6 |
| F4 | 0.59 | 17.2 | 11,880.1 | 10,614.6 |
| F5 | 0.66 | 12.4 | 5909.9 | 5791.4 |
| F6 | 0.67 | 20.6 | 4957.8 | 4606.5 |

**Table 3.** Questionnaire of the thermal sensation scale.

| Description | | | | | | | |
|---|---|---|---|---|---|---|---|
| Q1 | Q2 | Q3 | Q4 | Q5 | Q6 | Sensation | Score |
| Adhering to body | Hot | Humid | Impervious | Stiff | Uncomfortable overall | Very | −3 |
| | | | | | | Comparatively | −2 |
| | | | | | | Slightly | −1 |
| Neutral | Neutral | Neutral | Neutral | Neutral | Neutral | Neutral | 0 |
| | | | | | | Slightly | +1 |
| | | | | | | Comparatively | +2 |
| Smooth | Cool | Dry | Permeable | Flexible | Comfortable overall | Very | +3 |

**Table 5.** Water loss rates of the lining combinations under simulated hydraulic pressures.

| Lining combination | Water depth (m) | Water loss rate at 3 min (%) | Water loss rate at 10 min (%) | Increase from 3 to 10 min (%) |
|---|---|---|---|---|
| F1 | 0.4 | 40.76 | 45.67 | 12.05 |
| | 0.7 | 45.42 | 49.81 | 9.67 |
| | 1 | 48.51 | 52.04 | 7.28 |
| | 1.3 | 51.04 | 57.44 | 12.54 |
| F2 | 0.4 | 46.29 | 50.65 | 9.42 |
| | 0.7 | 47.12 | 51.72 | 9.76 |
| | 1 | 50.06 | 53.74 | 7.35 |
| | 1.3 | 51.72 | 56.26 | 8.77 |
| F3 | 0.4 | 15.6 | 22.1 | 41.67 |
| | 0.7 | 25.29 | 30.24 | 19.57 |
| | 1 | 30.62 | 35.41 | 15.64 |
| | 1.3 | 31.94 | 36.23 | 13.43 |
| F4 | 0.4 | 50.09 | 54.72 | 9.24 |
| | 0.7 | 55.54 | 59.67 | 7.44 |
| | 1 | 57.57 | 61.95 | 7.61 |
| | 1.3 | 58.32 | 62.52 | 7.20 |
| F5 | 0.4 | 32.83 | 36.78 | 12.03 |
| | 0.7 | 34.75 | 38.74 | 11.48 |
| | 1 | 34.51 | 38.2 | 10.69 |
| | 1.3 | 48.5 | 52.63 | 8.52 |
| F6 | 0.4 | 47.06 | 52.42 | 11.39 |
| | 0.7 | 52.16 | 56.15 | 7.65 |
| | 1 | 55.14 | 59.44 | 7.80 |
| | 1.3 | 56.71 | 60.87 | 7.34 |

1.3 m, and the largest increase (41.67%) from 3 to 10 minutes when the hydraulic pressure was 0.4 m. With the increase of the hydraulic pressure, the water loss rates of the six samples at 3 minutes increased more rapidly than at 10 minutes. This means that under all the hydraulic pressures, the water was transferred quickly from the fabric to the water-absorbent resin in a short period of time (3 minutes), and this effect lessened when the time increased to 10 minutes. Under 1.3 m hydraulic pressure, the water transfer ability of F1 and F2 at 10 minutes slightly increased, as compared with that at 3 minutes. Their increases (12.54%, 8.77%) are higher than those for F4–F6 (7.20%, 8.52%, 7.34%), whose water loss rates remained at the same level.

Overall, the better the water transfer ability of the lining combination, the smaller the difference in water loss rate between the two time periods.

Because some of these fabrics exhibited >10% increases in the water loss rate from 3 to 10 minutes, we adopted the data at 10 minutes in the analysis. Figures 5(a) and (b) show the water loss rate and the water absorption rate of the six lining combination at 10 minutes under the four simulated hydraulic pressures. For all combinations under all four pressures, the water loss rates were higher than the water absorption rate of the water-absorbent resin materials, indicating that there were small losses during the water transfer. Lining combination F4 had the largest water loss/absorption ratio due to having the strongest water exchange ability, while lining combination F3 had the smallest. Considering the fabric hygroscopicity test result, it is demonstrated that F3 has the best water

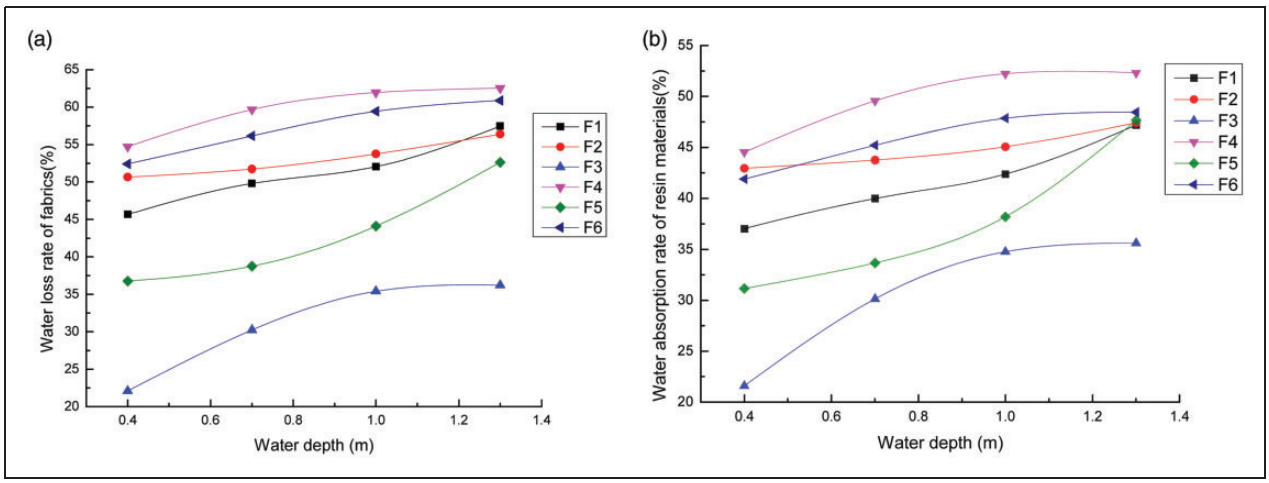


**Figure 5.** Water loss rate and water absorption rate of the six lining combinations at 10 minutes: (a) water loss rate of the fabric; (b) water absorption rate of resin materials.

absorption and retention. F3 also showed the fastest increase in water loss/absorption rates in the lower water depth range (0.4–1 m), while F5 demonstrated sharper rises in the higher water depth range (1–1.3 m). This was consistent with the conclusion that the hygroscopicity of a bottom fabric is higher than that of a top fabric because of the effect of gravity[19] in the fabric test.

## Comfort evaluation

*Objective evaluation.* Figure 6 displays the average temperatures and RH measured at the four body parts of the 10 subjects wearing the two chest waders. T-cl and RH-cl denote the temperature and the RH of the common chest wader, respectively. T-nl and RH-nl are the temperature and the RH of the new lining chest wader, respectively.

Both chest waders helped the body temperatures to maintain in a relatively small range of 30–35°C, with the temperature of the new lining chest wader being slightly higher and more stable than that of the common chest wader. This should be attributed to the fact that the new chest wader is thicker. The change of average RH (RH-cl and RH-nl) of the two chest waders could be divided into three stages. During 0–25 minutes, RH-cl remained lower than RH-nl. RH-cl was nearly unchanged within the first 10 minutes, and then increased rapidly from 10–25 minutes. During 25–37 minutes, RH-cl and RH-nl had consistent increase rates. During 37–60 minutes, RH-nl became substantially lower than RH-cl. After 37 minutes, RH-nl fluctuated only in a range of 47–50%, as opposed to the 50–70% range of RH-cl. Compared to the common chest wader, the new chest wader could help to lower the RH by >15% and to keep RH-nl at a level near 50%.

In the first half of the test, subjects mainly discharged insensible perspiration. There was no apparent difference between the common and new lining waders. In the second half of the test, a large amount of vapor and perspiration were generated because of continuous physiological metabolism and physical movements. The hybrid lining demonstrated an advantage in absorption and transfer of water. The RH of the new lining wader reached a stable state at about 50%, while that of the common chest wader reached a stable state at 68%.

Figure 7 provides comparisons of the average and the standard deviations of temperatures measured at the four body parts when 10 subjects wore the two chest waders. At the four parts, the temperatures measured with the new lining chest wader were all higher than those with the common chest wader. Via one-way analysis of variance (ANOVA), it was found that at the confidence level of $p > 0.05$, the temperatures at the calf, crotch and thigh were significantly different, while the temperatures at the back were not significantly different. The standard deviations of the two sets of temperatures were all small (<0.7).

Figure 8 shows the differences in the average RH at the four parts between the two chest waders. Except at

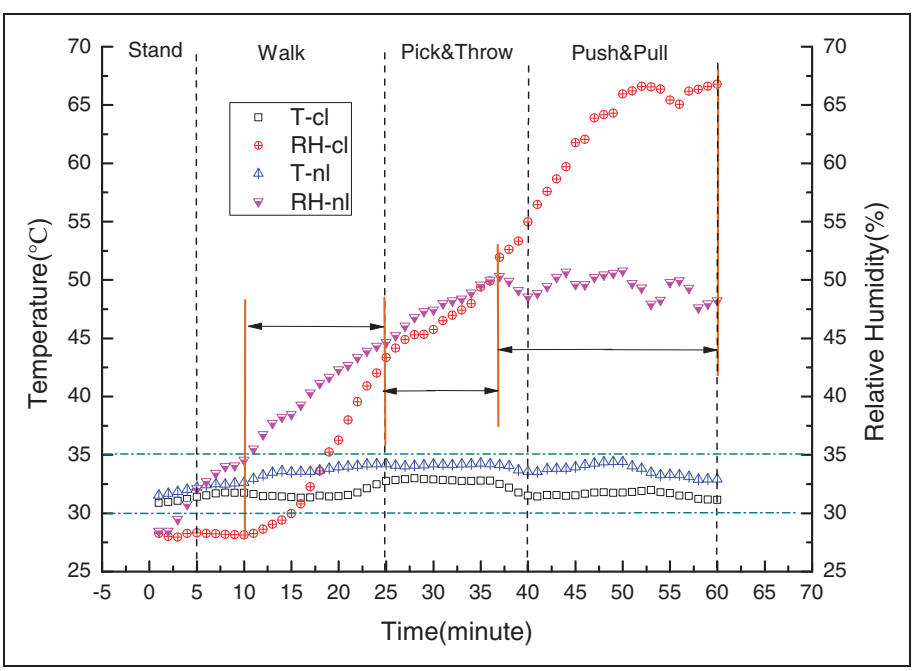


**Figure 6.** Average temperature and relative humidity at the four body parts of subjects wearing the two chest waders.

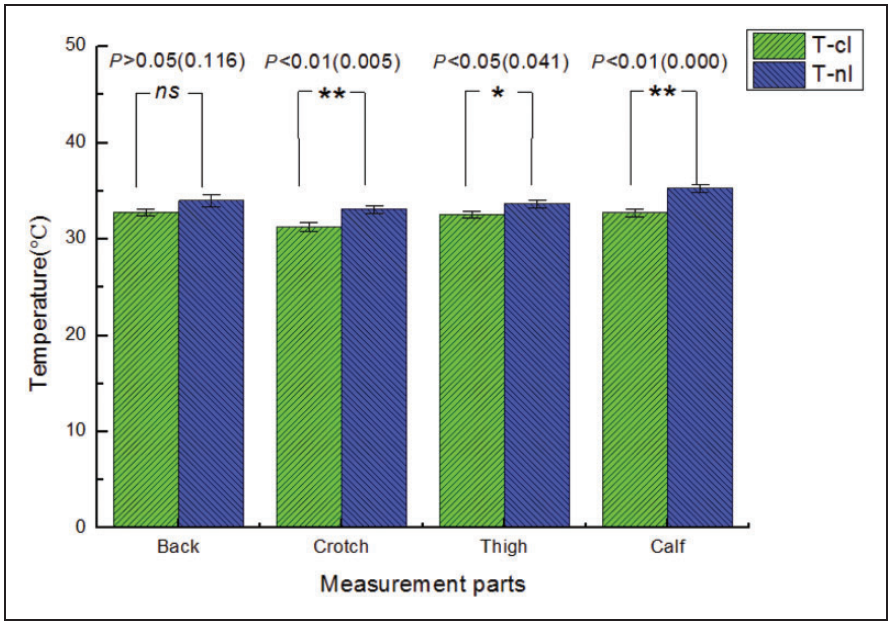


**Figure 7.** Comparison of temperatures at the four parts. *ns*: significant at $p > 0.05$; *significant at $p < 0.05$; **significant at $p < 0.01$

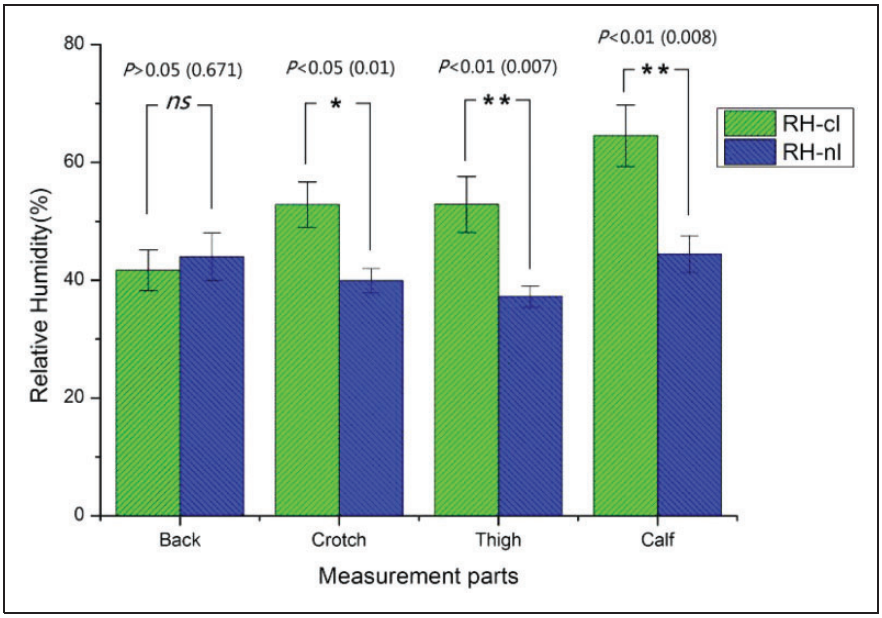


**Figure 8.** Comparison of relative humidity at the four parts. *ns*: significant at $p > 0.05$; *significant at $p < 0.05$; **significant at $p < 0.01$.

the back, the new lining chest wader had significantly lower humidity at the other three parts compared with the common chest wader. Specifically, the humidity was reduced by 20.1% at the calf, 15.6% at the thigh and 12.9% at the crotch. Because the back position was close to the upper hem of the chest wader, which had a large opening, the ventilation was more effective than for the other body parts during the whole test. This is why the data at the back appeared to be consistent with the environment. Thus, the differences in both temperature and RH between the new lining chest wader and the common one at the back were insignificant.

Through one-way ANVOA on the two groups, the difference at the crotch was significant at $p < 0.05$, and

the differences at the thigh and calf were significant at $p < 0.01$. Thus, it is necessary to analyze the RH differences at these three parts.

Figures 9–11 exhibit the changes in RH at the crotch, thigh and calf over time. During the first 36 and 42 minutes respectively, the humidity at the crotch and thigh of the new lining chest wader (RH-nl) were higher than that of the common chest wader (RH-cl), and became lower after 36 and 42 minutes, respectively. At the calf, RH-nl appeared to be lower than RH-cl throughout the entire test procedure, with the reduction being 90% at the end of the test.

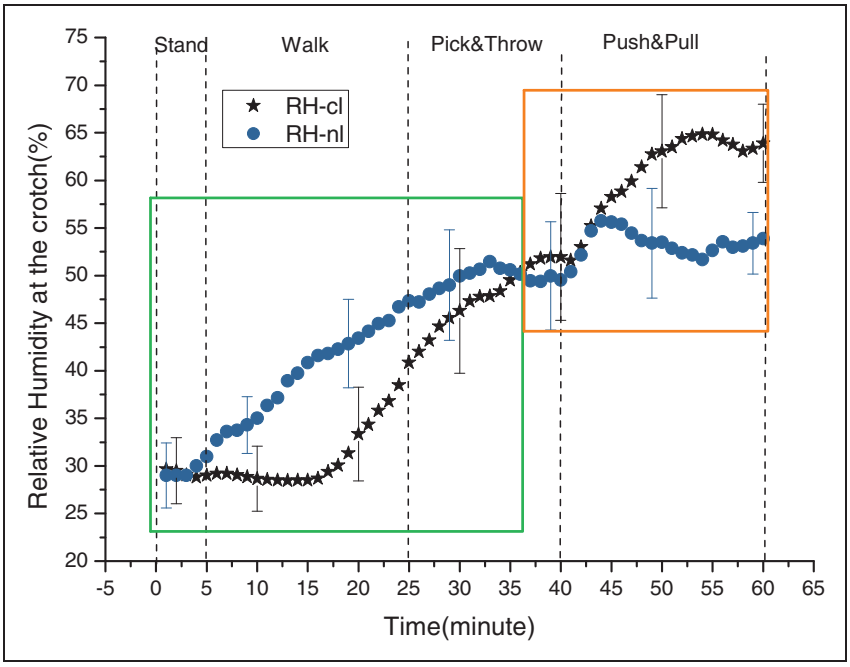


**Figure 9.** Relative humidity at the crotch.

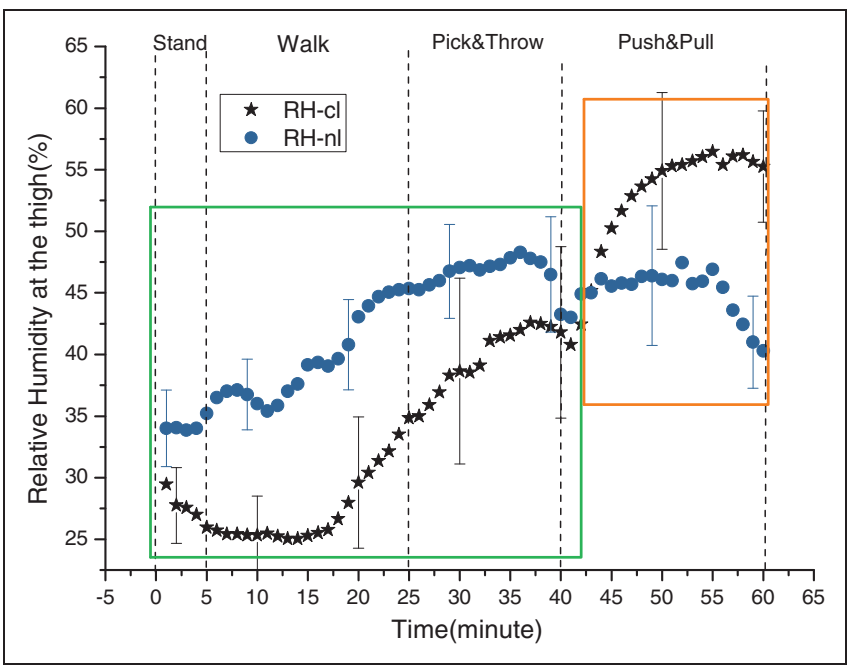


**Figure 10.** Relative humidity at the thigh.

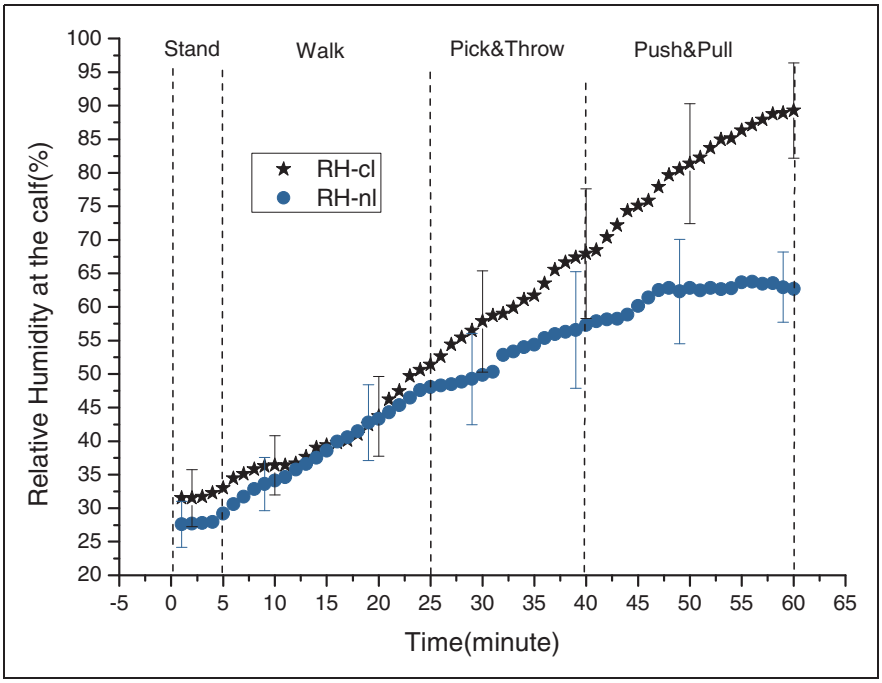


**Figure 11.** Relative humidity at the calf.

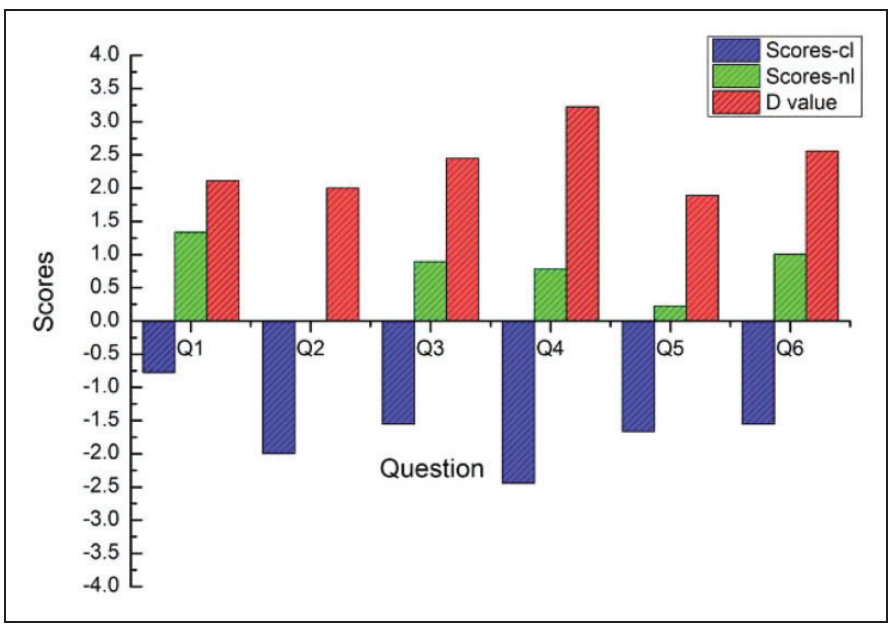


**Figure 12.** Subjective evaluation of the comfort of the chest wader.

The reason for this was that the structure of the common chest wader was loose, and would not fit tightly with the body at the thigh and crotch. By contrast, the ease allowance at the calf was $7 \pm 3$ cm, far smaller than those at the thigh and the crotch. The frequency and range of motion at the calf were higher than at the other parts. The small ease allowance at the calf made the new lining have better humidity absorption and conductivity performance.

From a gender perspective, the female subjects had slightly lower average RH at the four body parts (5.43%) than the male subjects at the end of the test. This means that the males perspired more heavily than the females at the given operation strength of the test.

*Subjective evaluation.* The subjective evaluation results from the subjects are shown in Figure 12, in which *D* values are the rating differences between the two chest waders. It can be seen that the scores for the new lining chest wader were higher than for the common chest wader over all six questions in Table 3. This means that the new lining has comprehensively improved the thermal-moisture comfort of the chest wader, especially for permeability (Q4), overall comfort (Q6) and humidity (Q3). This was consistent with the conclusion in a previous paper.[8]

The new lining was better than the common chest wader in skin feel. As the results of Q1 and Q2 show, the "adhering to body" and "hot" sensations declined when the subjects wore the new lining chest wader. On the aspect of "activity convenience" (Q5), the new lining chest wader improved the least because the lining pattern used the same dimensions, except for the crotch.

The statistics of the questionnaire answers also indicated that the calf was the hottest, stuffiest and moistest part, regardless of the type of chest wader, which was consistent with the RH test result shown in Figure 11. These chest waders were relatively tight on the lower portion and loose on the upper portion, and therefore the generated sweat at the feet and calves was harder to discharge in a timely manner. The excessive impermeability and moisture caused a discomfort feeling even with the new lining. Regarding the comfort feelings at the other parts, the subjects noticed the differences between the two chest waders: 33.3% at the crotch and thigh, 22.2% at the calf and 11.1% at the back. The females gave similar results as the males for the most sensation questions when comparing the new lining chest wader with the common one. Nevertheless, the difference values of the females were higher than the males for Q1 and Q3, indicating that the female sensation on the skin feel was more sensitive than for the males.

## Conclusions

This paper presented a hybrid design of a new lining chest wader for the fishing industry. The new chest wader utilized different lining materials for different body regions to meet the body sweating pattern. The moisture permeability tests showed that the hygroscopicity and moisture conductivity of the fabrics would change with hydraulic pressure alterations. Therefore, this should be taken into consideration when carrying out the performance test for selecting chest wader fabrics. The overall comfort evaluation tests demonstrated that the new lining could improve the degree of comfort of the chest wader significantly. The average RH of the new lining chest wader can be reduced by about a third after 60 minutes. As the hydraulic pressure increased, the improvement of moisture comfort was enhanced most at the calf, followed by that at the thigh and crotch. In the subjective evaluation, most of the participants indicated significant comfort improvements at the crotch and thigh. This inconsistency between the two evaluation methods should be taken into account in the future design of the thermal-moisture comfort of the chest wader.

### Declaration of Conflicting Interests

The authors declared no potential conflicts of interest with respect to the research, authorship and/or publication of this article.

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